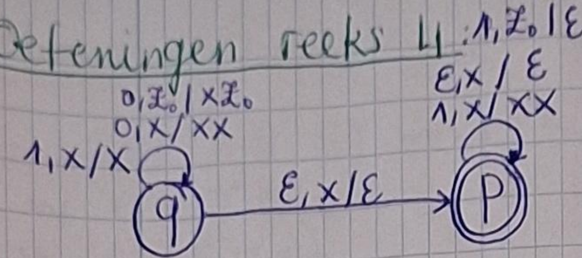
* Oefeningen reeks 4

1.

a)



b)

(a) $(q, 01, Z_0)$

$(q, 1, xZ_0)$

(q, ε, xZ_0) $\quad$ $(p, 1, Z_0)$

(p, ε, Z_0) $\quad$ $(p, \varepsilon, \varepsilon)$

(b)

$(q, 0011, Z_0)$

$(q, 011, xZ_0)$

$(q, 11, xxZ_0)$ $\quad$ $(p, 011, xZ_0)$

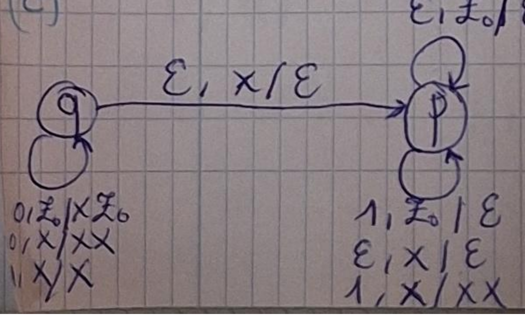
$(q, 1, xxZ_0)$ $\quad$ $(p, 11, xxZ_0)$

$(q, \varepsilon, xx\cancel{Z_0})$ $\quad$ $(p, 1, xxZ_0)$ $\quad$ $(p, 1, xxx\cancel{Z_0})$

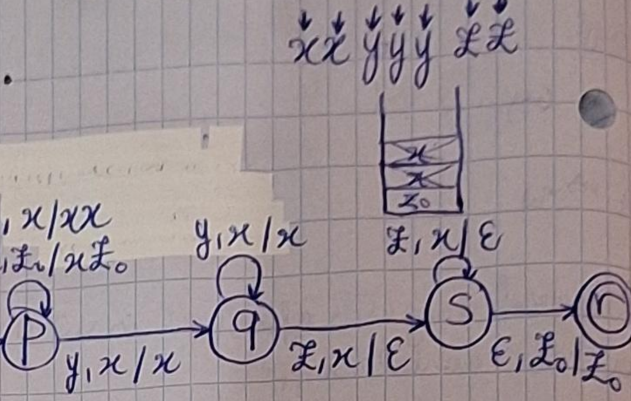
$(p, \varepsilon, x\cancel{Z_0})$ $\quad$ $(p, \varepsilon, xxxZ_0)$ $\quad$ $(p, \varepsilon, \cancel{xxxZ_0})$

(p, ε, Z_0)

(c)



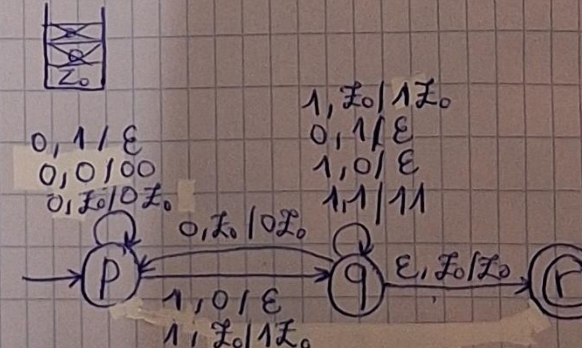
2.



$\delta(p, x, Z_0) = (p, xZ_0)$
$\delta(p, x, x) = (p, xx)$
$\delta(p, y, x) = (q, x)$
$\delta(q, y, x) = (p, x)$
$\delta(q, z, x) = (s, \varepsilon)$

$P = (\{p, q, r, s\}, \{x, y, z\}, \{x, Z_0\}, \delta, p, Z_0, \{r\})$

3. $\quad$ 0101 $\quad$ 0011
$\quad$ 1010 $\quad$ 1100

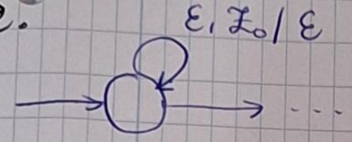


$\delta(p, 0, Z_0) = (p, 0Z_0)$
$\delta(p, 1, 0) = (q, \varepsilon)$
δ
$\vdots$

$P = (\{p, q, r\}, \{0, 1\}, \{0, 1, Z_0\}, \delta, p, Z_0, \{r\})$

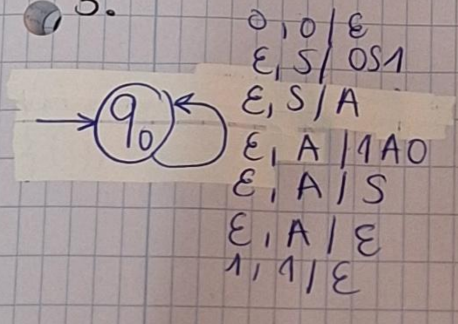
* Oefeningen reeks 5

2.

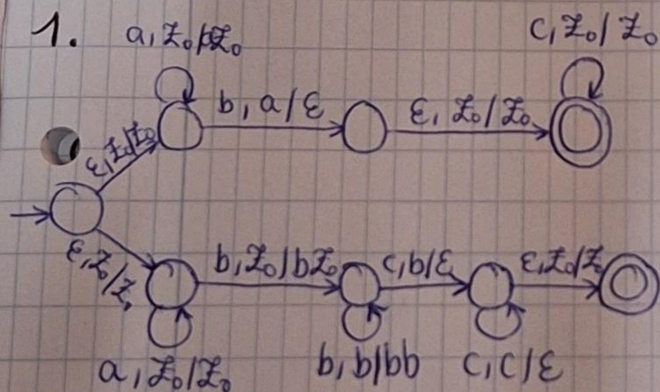


We kunnen dat doen door de transitie ε, Z_0 / ε toe te voegen aan de start state.

3.

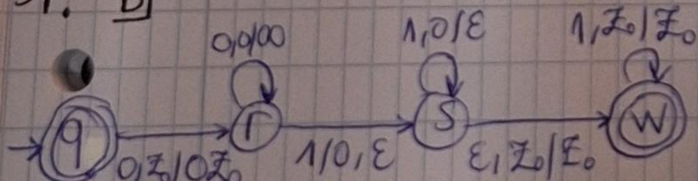


1.

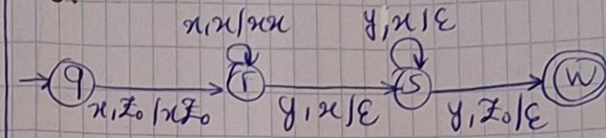


* Oefeningen reeks 6

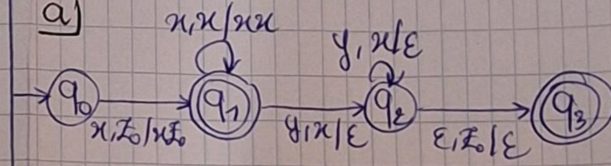
1. b)



2.



3.
a)



* Oefeningen reeks 11

1. b. (b)

$(q, 0011, Z_0)$

$(q, 011, XZ_0)$

$(q, 11, XXZ_0)$ $(p, 011, Z_0)$

$(q, 1, XXZ_0)$ $(p, 11, XZ_0)$

(q, ε, XXZ_0) $(p, 1, XZ_0)$ $(p, 1, XXZ_0)$ $(p, 11, Z_0)$

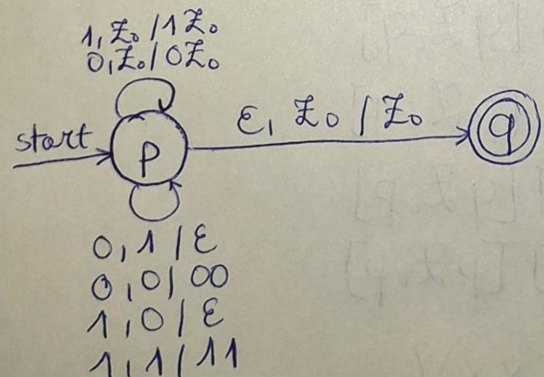
(p, ε, XZ_0) (p, ε, XXZ_0) $(p, 1, Z_0)$ $(p, \varepsilon, XXXZ_0)$ $(p, 1, XZ_0)$ $(p, 1, \varepsilon$

(p, ε, Z_0) (p, ε, XZ_0) $(p, \varepsilon, \varepsilon)$ (p, ε, XXZ_0) (p, ε, XXZ_0) $(p, 1, Z_0)$

(p, ε, Z_0) (p, ε, XZ_0) (p, ε, XZ_0) $(p, \varepsilon, \varepsilon$

(p, ε, Z_0) (p, ε, Z_0)

3.



* Oefeningen reeks 5

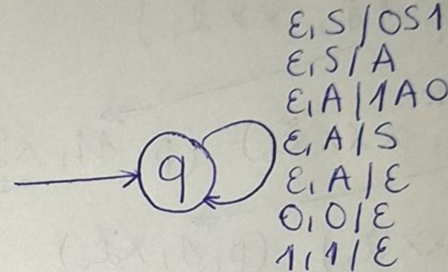
3. PDA, $P = (\{q\}, \{0, 1\}, \{S, A, 0, 1\}, \delta, q, S)$

$\delta(q, \varepsilon, S) = \{(q, 0S1), (q, A)\}$

$\delta(q, \varepsilon, A) = \{(q, 1A0), (q, S), (q, \varepsilon)\}$

$\delta(q, 0, 0) = \{(q, \varepsilon)\}$

$\delta(q, 1, 1) = \{(q, \varepsilon)\}$



4. CFG, $G = (V, \{0, 1\}, R, S)$

$V = \{S, [pXp], [pXq], [qXp], [qXq], [pZ_0p], [pZ_0q], [qZ_0p], [qZ_0q]\}$

(a) $\delta(q, 1, Z_0) = \{(q, XZ_0)\}$

$[qZ_0q] \rightarrow 1\,[qXq][qZ_0q]$

$[qZ_0q] \rightarrow 1\,[qXp][pZ_0q]$

$[qZ_0p] \rightarrow 1\,[qXq][qZ_0p]$

$[qZ_0p] \rightarrow 1\,[qXp][pZ_0p]$

(b) $\delta(q, 1, X) = \{(q, XX)\}$

$[qXq] \rightarrow 1\,[qXq][qXq]$

$[qXq] \rightarrow 1\,[qXp][pXq]$

$[qXp] \rightarrow 1\,[qXq][qXp]$

$[qXp] \rightarrow 1\,[qXp][pXp]$

(c) $\delta(q, 0, X) = \{(p, X)\}$

$[qXq] \rightarrow 0\,[pXq]$

$[qXp] \rightarrow 0\,[pXp]$

(d) $\delta(q, \varepsilon, X) = \{(q, \varepsilon)\}$

$[qXq] \longrightarrow \varepsilon$

(e) $\delta(p, 1, X) = \{(p, \varepsilon)\}$

$[pXp] \longrightarrow 1$

(f) $\delta(p, 0, Z_0) = \{(q, Z_0)\}$

$[pZ_0q] \longrightarrow 0\,[qZ_0q]$

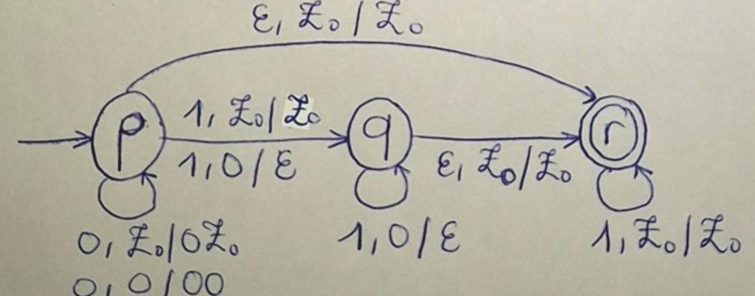
$[pZ_0p] \longrightarrow 0\,[qZ_0p]$

* Oefeningen reeks 6

1.

a. De PDA P is niet deterministisch want we hebben een keuze bij de staat q als de top van stack X is en de input we krijgen is 1.

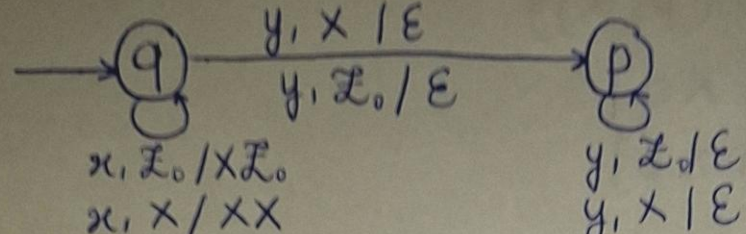
b.



~~2. $P = (\{q\}, \{x, y\}, \{S, x, y\}, \delta, q, S)$~~

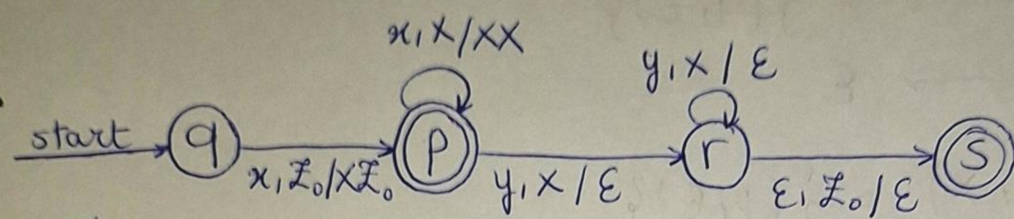
~~$\delta(q, \varepsilon, S) =$~~

2.



3.

a.



b. Nee, want we zullen niet kunnen de X'en poppen van de stack zodat we hetzelfde aantal y'en hebben voor de taal $\{x^n y^n \mid n \in \mathbb{N}_0\}$